Reiwa 5 (2023)

Date submitted: 20/11/2023

Fundamental Labs I Report

(School of Science and Engineering, College of Engineering Systems)

☐ 7 System control engineering basic labs
☐ 9 Diodes and transistors
☐ 11 DC motor manufacturing and control
☑ 8 Basics of linear systems using operational amplifier
☐ 10 Basics of logic circuits and computers
☐ 12 Mechanisms and mechanical elements

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Lab partners: (1) Nina Wells (2) Tian Vatcharapreecha
(3)_____ (4)_____
(5)_____ (6)_____

	1st week	2nd week	Report submission
Lab date	Month/Day	Month/Day	Received stamp
Lab participant's stamp with seal Non-participants leave blank	工学システム学類 基礎実験 23.11.07 22-1s 線形システム	工学システム学類 基礎実験 23.11.13 22-1s 線形システム	
Latecomer arrival time		12:18	

Use this sheet as the report cover page.
Reports are submitted by the date/time and to the location designated for each theme.
Submission deadline: Month: 11 Date: 20 (), Time: 12:15, Submission location: Manaba
NOTE: This should be collected and saved by each teaching assistant (TA) after the mandatory information has been initially filled in during the first lab. After their participation in the second lab has been understood, this sheet is returned to students who are instructed to use it as the cover page of their report.

Chapter 8: Basics of Linear Systems Using Operational Amplifiers

Week 1 Assignments

1 Purpose

An operational amplifier is an important active circuit element that is used for analog amplification and analog operation in electric circuits. The obective of this lab is to understand the operating characteristics of operational amplifier circuits by learning about and building analog basic amplifier circuits and analog operation circuits, such as those used for addition, subtraction, and calculus.

2 Fig. 4-1 (Inverting Amplifier Circuit)

2.1 Objective

To build and understand the workings of an inverting amplifier circuit.

2.2 Experiment Details

The circuit as shown in Fig 4-1 was configured and a direct current (DC) signal was used for input. In the first task, two resistors, 10kΩ and 100kΩ, are connected in series and parallel respectively to the operational amplifier which is linked to a 9V battery source and a DC current. Fig 1 shows the schematic diagram of the setup.

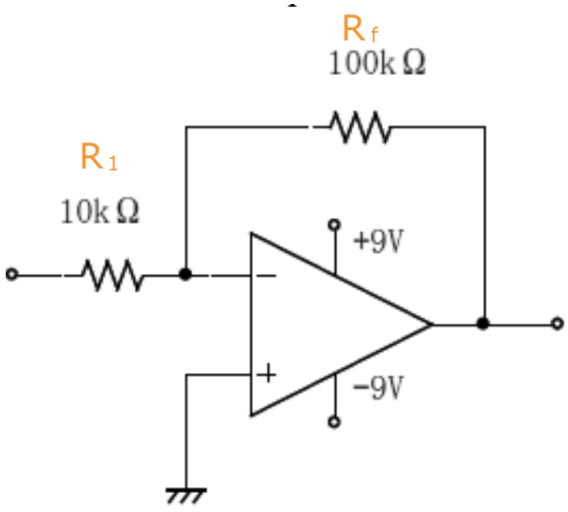


Figure 1: Inverting Amplifier Circuit

2.3 Results

Upon applying a direct current ranging from -1 V to 1 V with constant increments of 0.2 V to the circuit, the following results were observed. The oscilloscope showed a straight line (constant observed output voltage) of a different value for each increment, ranging from 7.5 V to -6 V. The values of the output voltage versus the input voltage is shown in Table 2, plotted as points in Graph 1[1].

2.4 Discussion

The circuit configured in this task resembles a standard inverting amplifier circuit. The gain (G_{4-1}) is given by,

$$G_{4-1} = -R_f/R_1 = -100k\Omega/10k\Omega = -10$$

Changing the input voltage (v_1) from -1V to 1V, the output voltage should theoretically change as follows,

$$v_2 = Gv_1 = (-10)v_1$$

This would give the results in Table 1.

Table 1: Theoretical Input-Output Voltage Values

Theoretical Input Voltage (V)	-1	-0.8	-0.6	-0.4	-0.2	0	0.2	0.4	0.6	0.8	1
Theoretical Output Voltage (V)	-10	-8	-6	-4	-2	0	2	4	6	8	10

However, since the circuit created has a DC Voltage input of 9V, this would saturate the maximum voltage value to around 9V. The observed results do not match this as the highest value is about 7.5 Volts. This is probably due to the internal resistance of the battery, it's power efficiency and other resistances of wires which would suggest that the batteries provide an effective voltage of around 7.5 Volts. An explanation for why the negative maximum output voltage was lower than the positive maximum output is perhaps a voltage offset in the output due to imperfections in the amplifier or other circuit components.

3 Fig. 4-2 (Integrator Circuit)

3.1 Objective

To build and understand the workings of an integrator circuit.

3.2 Experiment Details

The circuit as shown in Fig 4-2 was configured. Input was a square wave of alternating current (AC) about 200 Hz. In the second task, one 10kΩ resistor and one 0.1 μF, are

[1]Graphs 1-9 are attached at the end of the Appendix as they were hand-drawn

connected in series and parallel respectively to the operational amplifier which is linked to a 9V battery source, with an AC input. Fig 2 shows the schematic diagram of the setup.

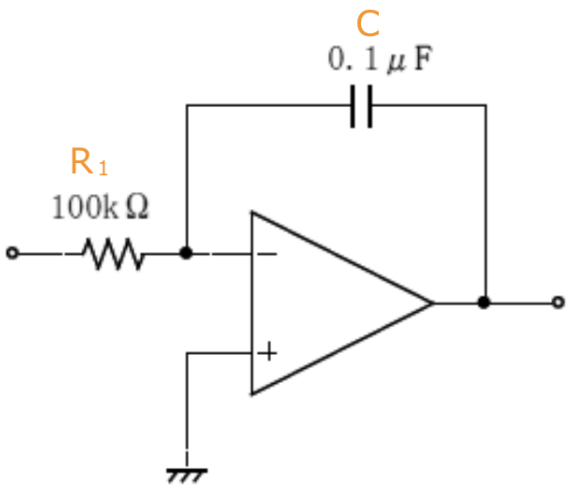


Figure 2: Integrator Circuit

3.3 Results

Upon applying an input square wave of 200 Hz, the continuous wave output of the oscilloscope was paused to show a Voltage-time graph identical to that shown in Graph 2. The Voltage-time graph of the output resembles a triangular wave whose crests coincide with the beginning of a crests of the square wave and the troughs of the triangular wave coincide with the end of a crest and the beginning of a trough in the square wave. The amplitudes of the triangular wave is seen to be slightly increasing with time.

3.4 Discussion

The circuit configured in this task resembles an integrator circuit. In an integrator circuit, the output voltage (V_out) can be calculated using the following formula [5]:

$$V_{out} = -\frac{1}{R_1 C}\int_0^t V_{in}\,dt = -\int_0^t V_{in}\frac{dt}{R_1 C}$$

The gain factor(G_{4-2}) is is thus given by,

$$G_{4-2} = -\frac{1}{R_1 C}$$

with respect to time.
In this circuit with $R_1 = 100k\Omega$ and $C = 0.1\mu F$, we get the magnitude of gain (G_{4-2}) as,

$$G_{4-2} = V_{out}/V_{in} = -\frac{1}{(100\times10^3)(0.1\times10^{-6})} = -1000$$

An integrator circuit integrates the input wave and then inverts it. At the end of a crest of a square wave, the integral would have the maximum positive value. This value would linearly decrease as the square wave reaches a trough. This would output a triangular wave. Since the gain inverts the input voltage, this triangular wave corresponds to the negative integral of the input wave. There is a gradual increase observed in the maximum values of

the triangular output wave due to the gain in the integral, as opposed to a constant one. This could also be due to the voltage offset (a slight addition or subtraction from the output voltage due to imperfections in components of the amplifier).

4 Fig. 4-3 (Addition Circuit)

4.1 Objective

To build and understand the workings of an addition circuit.

4.2 Experiment Details

The circuit as shown in Fig 4-3 was configured. Direct current was used for input e_1 and approximately 1 kHz sine wave was used for input e_2. In the third task, two voltage sources e_1 and e_2 with an AC input and DC input respectively connected with two 10kω resistors that are connected in parallel to each other and in series to the amplifier in addition to another 10kΩ resistor connected in parallel to the amplifier. The amplifier is connected to a 9V battery source and ground. Fig 3 shows the schematic diagram of the setup.

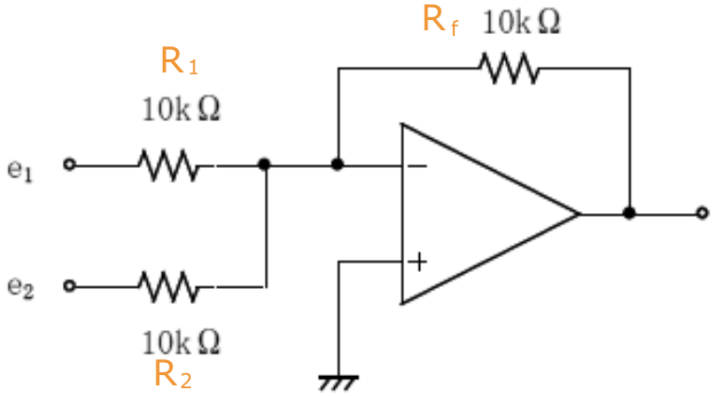


Figure 3: Addition Circuit

4.3 Results

When current was applied, the oscilloscope showed two sine waves, one of the input voltage and one of the output voltage. The output voltage corresponds to a reflection of the input voltage along the x-axis. This effectively results in a vertical translation downwards by 200 mV and a phase difference of π. The results are shown in Graph 3.

4.4 Discussion

The circuit configured in this task resembles an addition circuit. The gain (G_{4-3}) is given by,
For e_1,

$$G_{4-3}(e_1) = -R_f/R_1 = -101k\Omega/10k\Omega = -1$$

For e_2,

$$G_{4-3}(e_2) = -R_f/R_2 = -10k\Omega/10k\Omega = -1$$

An addition circuit adds a constant value to the circuit to result in a larger output voltage which is then inverted to provide an effectively lesser voltage. In the results observed with this circuit, (from Graph 3) it can be interpreted that the output wave was translated upwards by ≈800mV and then inverted. If this were an open-loop circuit (no inversion), the output voltage would be incredibly large (by 800mV), but due to the closed-loop circuit, the gain in output is effectively lesser by 200mV.

5 Fig. 4-4 (Subtraction Circuit)

5.1 Objective

To build and understand the workings of a subtraction circuit.

5.2 Experiment Details

The circuit as shown in Fig 4-2 was configured. Voltage was divided by variable resistance and a sine wave was inputted. A variable resistance dividing input voltage is added to the circuit. Fig 4 shows the schematic diagram of the setup.

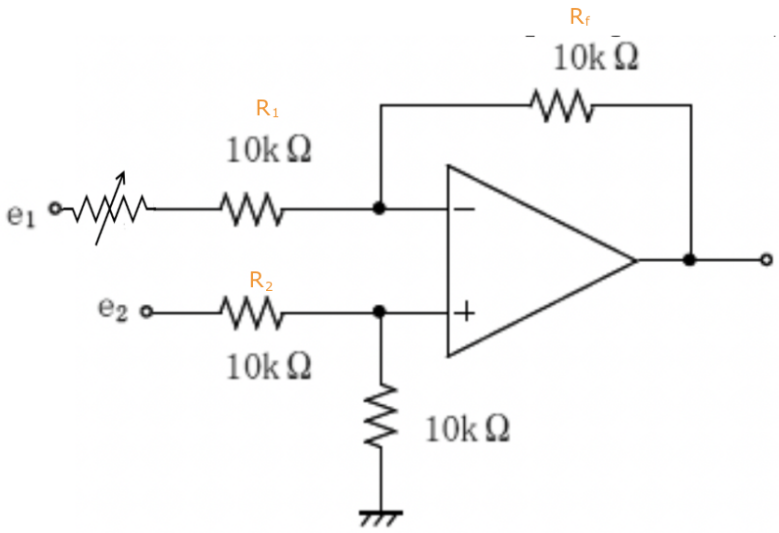


Figure 4: Subtraction Circuit

5.3 Results

When voltage was applied, two sine waves corresponding to the input and output voltage values can be seen on the oscilloscope. Depending on the change in variable resistance, the output voltage wave is stretched and compressed. When the variable resistance is decreased, the output voltage gets smaller (wave is compressed) until it resembles the input voltage wave as shown in Graph 4. When the variable resistance is increased, the output voltage is larger (wave is stretched) as shown in Graph 5.

5.4 Discussion

The circuit configured in this task resembles a subtraction circuit.The magnitude of gain (G_{4-4}) is given by,

$$G_{4-4} = \frac{R_f}{R_1} = 10/10 = 1$$

The output voltage (v_{out}) can thus be calculated using the following equation,

$$v_{out} = \frac{R_f}{R_1}(e_2 - e_1) = e_2 - e_1$$

Since the variable resistance is applied to e_1, if the variable resistance is increased, e_2 will decrease and the output voltage will increase. The opposite is also true (if the variable resistance is reduced, e_1 will increase and therefore the output voltage will decrease). The minimum value would therefore be $v_{out} = e_2$ which is the same as the input value. The maximum value would probably have a saturation point of less than 9V due to the battery limitations.

6 Fig. 4-5

6.1 Objective

To build and understand the workings of the circuit described in Figure 4-5 in the textbook.

6.2 Experiment Details

Fig 5 shows the experimental setup for this task. A capacitor (C = 0.1 μF) was connected in series to the operational amplifier and resistance (R = 100kΩ) was connected in parallel. The circuit was connected to an AC power source, a 9V battery and to ground.

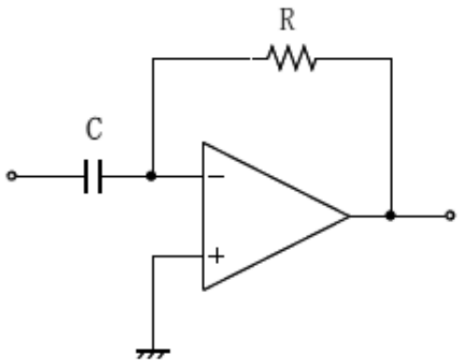


Figure 5: Differentiator Circuit

6.3 Results

When a continuous sine wave is inputted, the output voltage is a continuous sine wave translated by $\pi/4$ and compressed vertically to approximately one-half of the input amplitude as shown in Graph 6. When a triangular wave is inputted, the output resembles a square wave with noise at points corresponding to the peaks of the triangular wave as shown in Graph 7.

6.4 Discussion

This circuit can be defined as a differentiator circuit. The gain (G_{4-15}) is given by,

$$G_{4-5} = -R_f/X_c = -(100 \times 10^3)(2)(\pi)(5)(0.1 \times 10^{-6}) = -0.0314$$

where $X_c = 1/2\pi fC$ is the reactance of the capacitor depending on the input frequency. In this case, $f \approx 5Hz$ (judging from Graph 6 and 7).
The output wave is the derivative of the input wave. For a continuous sin wave input, the output is a cosine wave representing the value of the slope of the input wave. For a triangular wave input, there is oscillation observed at the values corresponding to the input peaks. This is because those points are non-differentiable, causing noise in the oscilloscope.

Week 2 Assignments

1 Purpose

The purpose of this week was to focus on the basic principles and characteristics of contact position measurement using a contact position sensor, an amplifier circuit using pressure-voltage conversion and an operational amplifier, and the principle and application of analog-digital conversion.

2 Assignment 1

In Assignment 1, a SoftPot contact sensor was used to detect the position pressed by a finger as accurately possible, a circuit was configured and the output voltage was measured using an oscilloscope. Figure 6 shows the schematic of the configured circuit. R_sensor refers to the resistance provided by the sensor and R_f was chosen to be 510ω to output a voltage that was ≈4.5V.

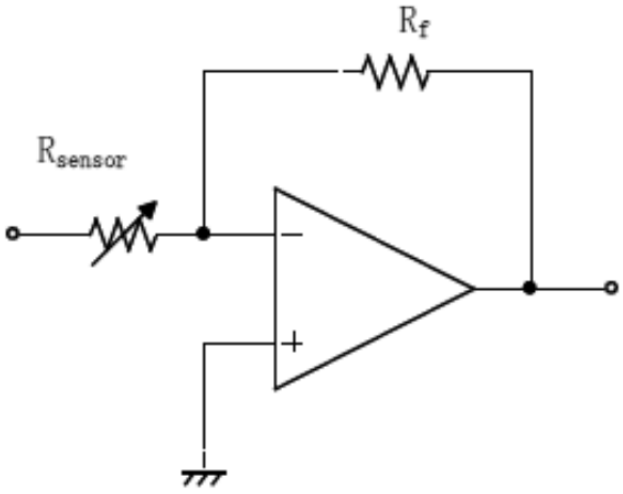


Figure 6: Assignment 1 Schematic

3 Assignment 2

3.1 Objective

The above circuit was used to determine the distance-voltage characteristics according to the AD-conversion board (Interface PCI-3174).

3.2 Experiment Details

A ruler and a pen was used to touch the sensor from a displacement of 0 cm from the tip to a displacement of 10 cm from it with increments of 0.5 cm and the Digital and Analog Voltage Data was recorded using the Analog-to-Digital and Digital-to-Analog functions in Matlab using Program 1[2]. The data was recorded.

3.3 Results

The data seems to have a positive non-linear trend in the increase in output voltage with distance. A quadic polynomial fit was determined for he distance-voltage characteristics data using Excel. These characteristics are described in Table 2 and in Graph 8.

Table 2: Distance Vs. Output Voltage

Distance (cm)	Output Voltage (V)
0	0.00
0.5	0.30
1	0.32
1.5	0.34
2	0.36
2.5	0.40
3	0.41
3.5	0.46
4	0.48
4.5	0.53
5	0.57
5.5	0.64
6	0.72
6.5	0.82
7	1.00
7.5	1.20
8	1.47
8.5	1.85
9	3.17
9.5	4.30

[2]The programs are listed in the Appendix

3.4 Discussion

The distance-voltage data can be best modelled using the following equation, where y is output voltage in volts and x is the distance from the tip in centimeters.

$$y = 0.166 + -0.0404x + 0.0394x^2 + -5.08 \times 10^{-3}x^3 + 1.95 \times 10^{-4}x^4$$

Since the contact sensor works on the voltage divider principle which utilizes two resistance strips to compare the potential difference between the touched and untouched strips. Since this measurement depends on resitance-voltage relationships, for several resistors, this relationship is not perfectly linear. This would suggest that the material of the sensor has a non-linear characteristic. Other reasons for non-linearity can be mechanical non-linearity, electrical contact issues etc.

4 Assignment 3

4.1 Objective

The objective of this assignment was to use a Digital rangefinder function in Matlab to theoretically determine the distance-voltage characteristics of the touch sensor using Matlab.

4.2 Experiment Details

The inverse of the polynomial fit created in Assignment two was inserted in Program 2 was used to calculate theoretical distance from the tip of the touch sensor and plotted in Table 3. These values were compared with the observed values in Table 2. The comparison was graphed in Graph 9 to showcase the relationship with the real distance on the x-axis and the measured distance on the y-axis.

Table 3: Real Distance Vs. Measured (Calculated) Distance

Real Distance (cm)	Measured Distance (cm)
0	0.0
0.5	2.0
1	2.3
1.5	2.4
2	2.6
2.5	2.8
3	3.1
3.5	3.4
4	3.7
4.5	4.0
5	4.4
5.5	4.9
6	5.4
6.5	5.9
7	6.6
7.5	7.4
8	8.2
8.5	8.9
9	9.0
9.5	10.0

4.3 Discussion

Since the quadic equation in Assignment 2 modelled the change in output voltage with distance, it's inverse, $x' = -1.37 + 12.7y + -5.3y^2 + 0.858y^3 + -0.0404y^4$ would give the value of distance (x') for the measured voltage (y). When the strip is touched at a real distance (x), the program generates an output voltage. The model equation is used to calculate the measured distance (x'). This would have several applications in calculating an unknown distance at which a sensor is touched based on the voltage it measures.
Theoretically, the real distance and the measured distance should be equal, and therefore, Graph 9 should be modelled by the equation y=x. However in experimental data, there are some discrepancies that can be seen. Not all data points lie on y=x. In fact most points either lie above or below the line, except 3, 9, 5 which have measured values equal to the real values as shown in Table 3. Looking at Table 3, it can be seen that values from 2cm to 7.5 cm have a shift of ≈ 0.5 cm from the y = x, suggesting the possibility of systemic error in specific regions. However, the linear regression line of best fit is still $y = 0.949 * x + 0.263 \approx y = x$, suggesting the data is quite accurate. Since there is not a large difference in the y-intercepts, a majority of the error is therefore, not systemic but rather random. The reasons for random errors can be as follows:

1. Inaccuracies in the AD conversion when measuring the initial Voltage data in Assignment 2. A pen was used to touch the sensor manually, therefore minute differences in pressure and/or position touched could sway the digital values calculated. Moreover, the strip was calibrated from Analog (-5V to +5V) to Digital (0 to 4095), starting at $0 \approx 2046$. The value of digital would keep changing minutely due to the difference and poor contact.

2. This causes the initial quadic fit to not pass through all the points. If the initial polynomial fit is not entirely accurate, the inverse would also not be a perfect model of the real distances, but rather a calculated interpolation of them.

3. Other experimental errors could be power fluctuations, dynamic response etc. Environmental factors such as temperature, wear and tear could have caused the working of the sensor (which relies on resistance) to worsen over time. It is possible that the sensor we used had some discrepancies of a similar kind.

5 Assignment 4

5.1 Program 3

In the saw function, a period (T) of 200 (corresponding to 5ms) is set in line 2. The for-loop repeats the discrete value generation for one period, creating a line y = mx and then repeating that continuously creating a saw wave. During this, x = nCount that changes from 0 to 200 and m = wMax/200, where xMax = 4095 in every iteration to generate a visibly continuous graph.

5.2 Program 4

In the sine wave function, we generated the sine wave $asin(bx) + c$, where the program sets the amplitude (a = 2046) to wMid = (wMax+wMid)/2 where wMax = 4095 and wMin = 0; sets the angular frequency (b = $2\pi/T$ = (2)(3.14)/200); sets the vertical shift (c = wMid

= 2046) because the calculations involve values only from 0V to 5V. The for-loop for 200 changes the value of nCount from 0 to 200 which generates a value for x = nCount in the sine function. The for-loop repeats the value generation for one period, generating a sine wave.

5.3 Program 5

In the rectangle wave function, a similar for-loop is executed. The if-statements generate a positive constant amplitude of wMax = 4095 for half a period and an amplitude of wMin = 0 for the second half for x = nCount = 0 to 200.

5.4 Program 6

In the triangle wave function, another if-statement is used to generate a function y = mx (similar to the saw-function) with a positive slope m = 2wMid/100 (where 2wMid = wMax = 4095) and the same function with a negative slope -m = -2wMid/100.

6 Assignment 5

6.1 Sampling

In the case of AD conversion, sampling refers to the capturing of analog signals (the amplitude of an analog continuous function) over fixed intervals of time known as the sampling period[2].

6.2 Sampling Frequency

Sampling frequency refers to the number of samples taken per unit time during the digitization of an analog signal. It is the inverse of the sampling period.

6.3 Sampling Theorem

The Sampling Theorem states that a signal can be exactly reproduced if it is sampled at a frequency F, where F is greater than twice the maximum frequency in the signal[1].

6.4 Quantization

In analog to digital conversion, the continuous analog range is divided into a fixed number of intervals and depending upon the measured value, it is assigned to the closest fixed value. Quantization is the process by which a sampled amplitude value is assigned one of a limited number of fixed values[4].

6.5 Quantization Errors

Resolution of quantization refers to the number of intervals assigned to a given analog data. Quantization error is a systemic uncertainty that occurs during the digitization of analog values because of the limited resolution of the conversion. It refers to the difference in the actual analog value to its corresponding DA conversion value[3].

References

[1] *An introduction to sampling theory*. URL: http://www2.egr.uh.edu/~glover/applets/Sampling/Sampling.html#:~:text=The%20Sampling%20Theorem%20states%20that,maximum%20frequency%20in%20the%20signal..

[2] *Data Acquisition*. URL: https://ccs.fau.edu/~bressler/EDU/STSA/Modules/II.pdf.

[3] *Quantization error*. URL: https://www.ni.com/docs/ja-JP/bundle/ni-daqmx/page/mxcncpts/quanterror.html.

[4] *Sampling of A/D converter*. URL: https://en-support.renesas.com/knowledgeBase/16978396#:~:text=The%20A%2FD%20converter%20captures,is%20called%20the%20sampling%20frequency)..

[5] Wayne Storr. *Op-amp integrator - the operational amplifier integrator*. Aug. 2022. URL: https://www.electronics-tutorials.ws/opamp/opamp_6.html.

Appendix

6.6 Entire Matlab Code

```
#include <windows. h>
#include "AD_func. h"
#include "DA_func. h"
ADDATA addata,
DADATA dadata,
WORD DA_data[256],
double AD_data, // AD converted data entered double Volt, // voltage
double Length, //length
int nCount, //Number of repetitions
WORD wMax= 0xfff, //Maximum value
WORD wMin=0, //Minimum value
WORD wMid= (wMax + wMin) / 2, //Medium value
//Edit the following from here!
//Third week: Assignment 2: AD conversion
void kadai1 (void) {
Volt = 0.00244*AD_data-5;
}
void kadai2 (void) {
Length = -1.37+12.7*Volt-5.3*(Volt*Volt)+0.858*(Volt*Volt*Volt)-0.0404*
 (Volt*Volt*Volt*Volt);
}
//Second week: Assignment 4 DA conversion
//saw tooth wave
void saw (void) {
for (nCount = 0, nCount < 200, nCount++) {
DA_data[nCount] = (WORD) nCount * wMax / 200, }
}
//Sine wave
void sin (void) {
for(nCount=0;nCount<=200;nCount++){
DA_data[nCount]=(wMid)*sin(nCount*6.28/200)+wMid;
}
}
//square wave
void rec (void) {
for(nCount=0;nCount<200;nCount++){
if(nCount<=100){
DA_data[nCount]=wMin;
}
else{
DA_data[nCount]=wMax;
}
}
}
//triangle wave
```

```
void tri (void) {
for(nCount=0;nCount<200;nCount++){
if(nCount<=100){
DA_data[nCount]=wMid*2*nCount/100;
}
else{
DA_data[nCount]=-wMid*2*nCount/100;
}
}
}
```

6.7 Program 1

```
    void kadai1 (void) {
Volt = 0.00244*AD_data-5;
}
```

6.8 Program 2

```
    void kadai2 (void) {
Length = -1.37+12.7*Volt-5.3*(Volt*Volt)+0.858*(Volt*Volt*Volt)-0.0404*
 (Volt*Volt*Volt*Volt);
}
```

6.9 Program 3

```
void saw (void) {
for (nCount = 0, nCount < 200, nCount++) {
DA_data[nCount] = (WORD) nCount * wMax / 200, }
}
```

6.10 Program 4

```
//Sine wave
void sin (void) {
for(nCount=0;nCount<=200;nCount++){
DA_data[nCount]=(wMid)*sin(nCount*6.28/200)+wMid;
}
}
```

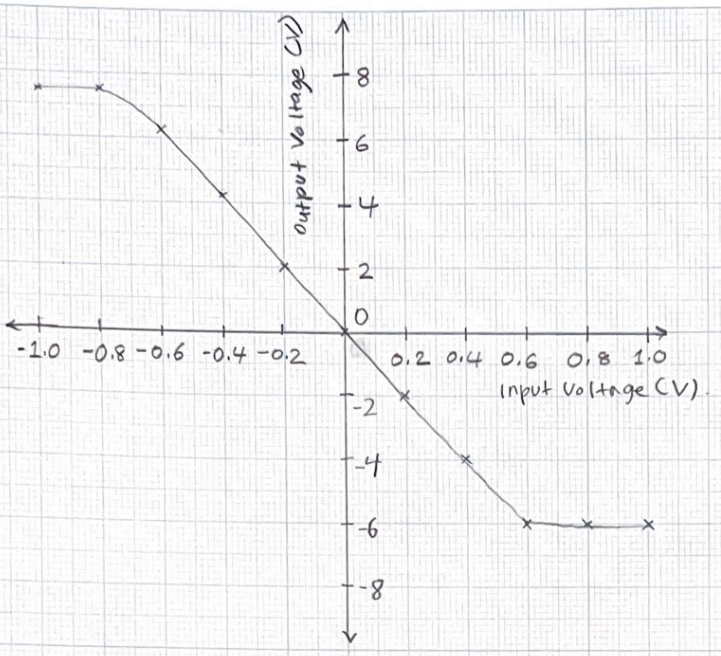
6.11 Program 5

```
//square wave
void rec (void) {
for(nCount=0;nCount<200;nCount++){
if(nCount<=100){
DA_data[nCount]=wMin;
}
```

```
else{
DA_data[nCount]=wMax;
}
}
}
```

6.12 Program 6

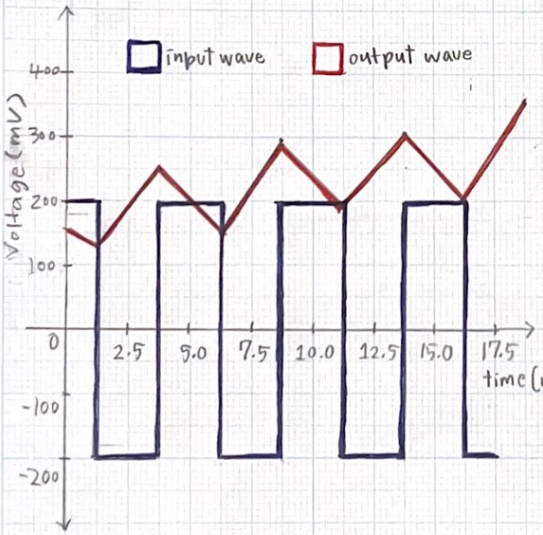
```
    //triangle wave
void tri (void) {
for(nCount=0;nCount<200;nCount++){
if(nCount<=100){
DA_data[nCount]=wMid*2*nCount/100;
}
else{
DA_data[nCount]=-wMid*2*nCount/100;
}
}
}
```



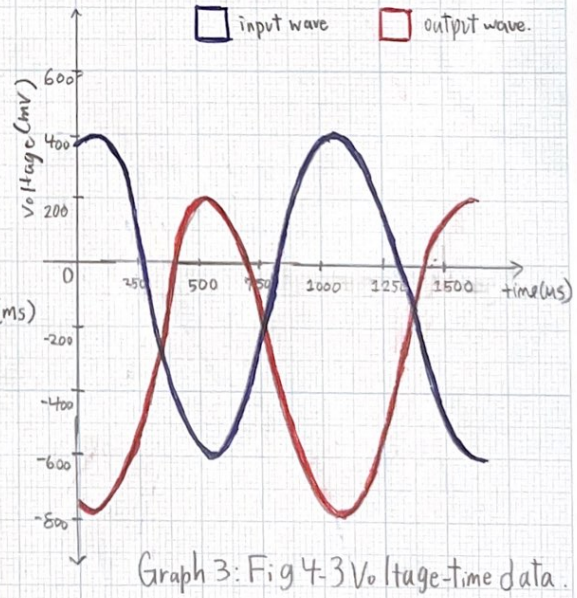
Graph 1: Fig 4-1 Input vs. Output Voltage Data.

Table 2: Fig4-1 Data

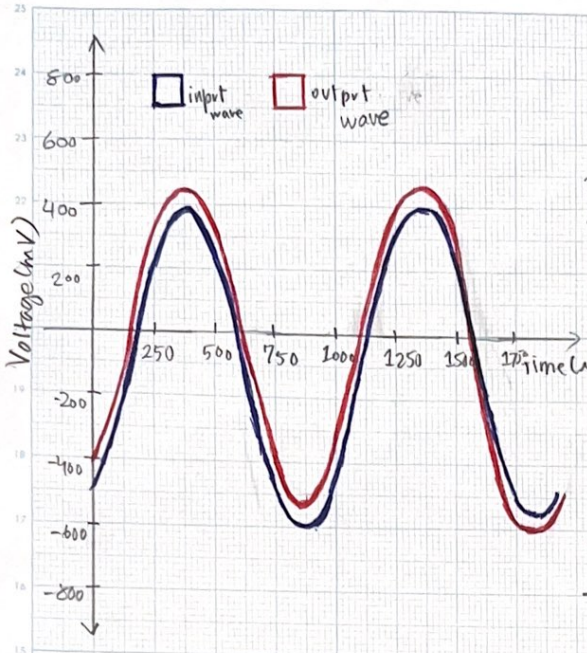
input voltage (V)	output voltage (V)
-1.0	7.5
-0.8	7.5
-0.6	6.25
-0.4	4.2
-0.2	2
0	0
0.2	-2
0.4	-4
0.6	-6
0.8	-6
1.0	-6



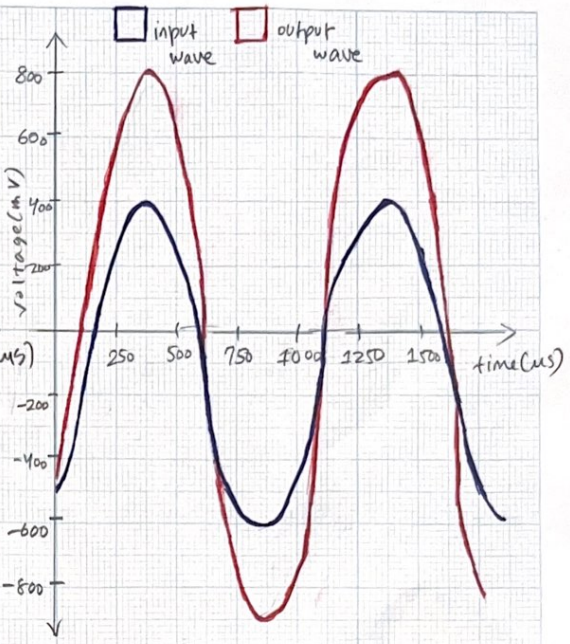
Graph 2: Fig4-2 Voltage-time data



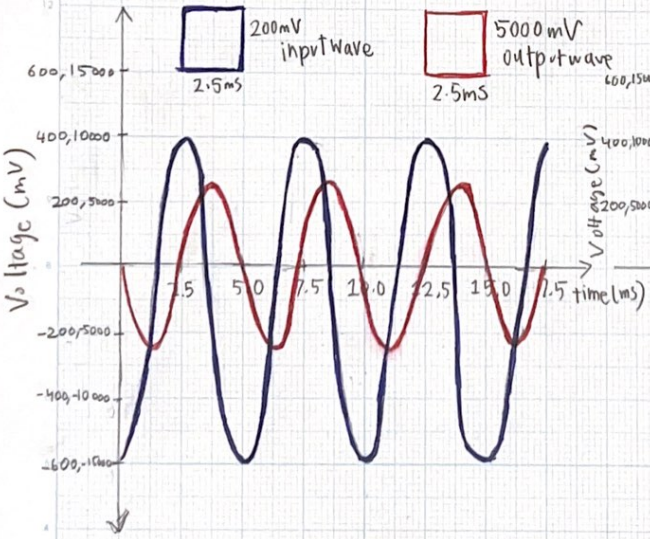
Graph 3: Fig 4-3 Voltage-time data.



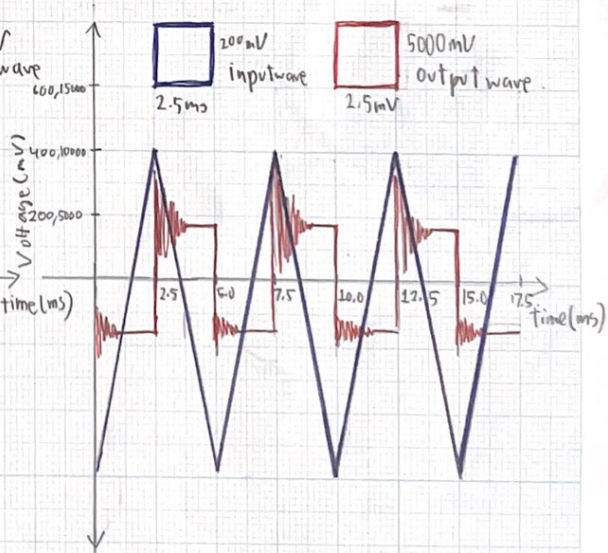
Graph 4: Fig 4-4 Voltage-time data for a high variable resistance



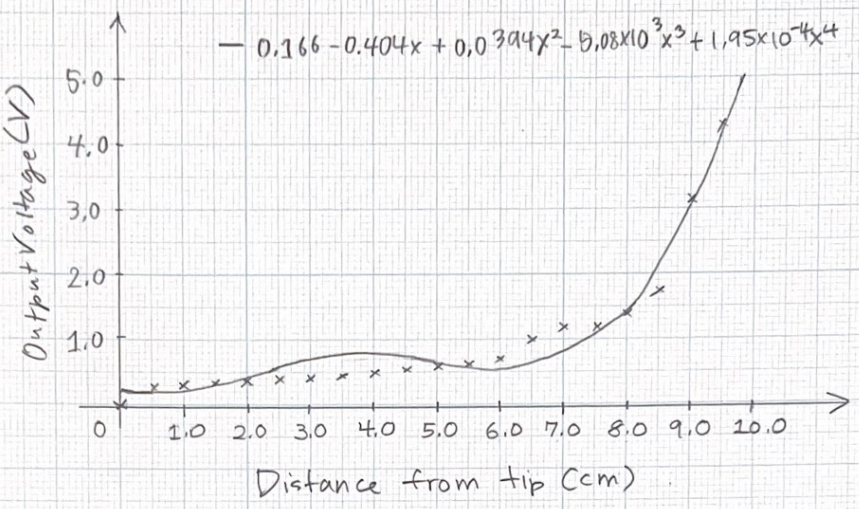
Graph 5: Fig 4-4 Voltage-time data for a low variable resistance



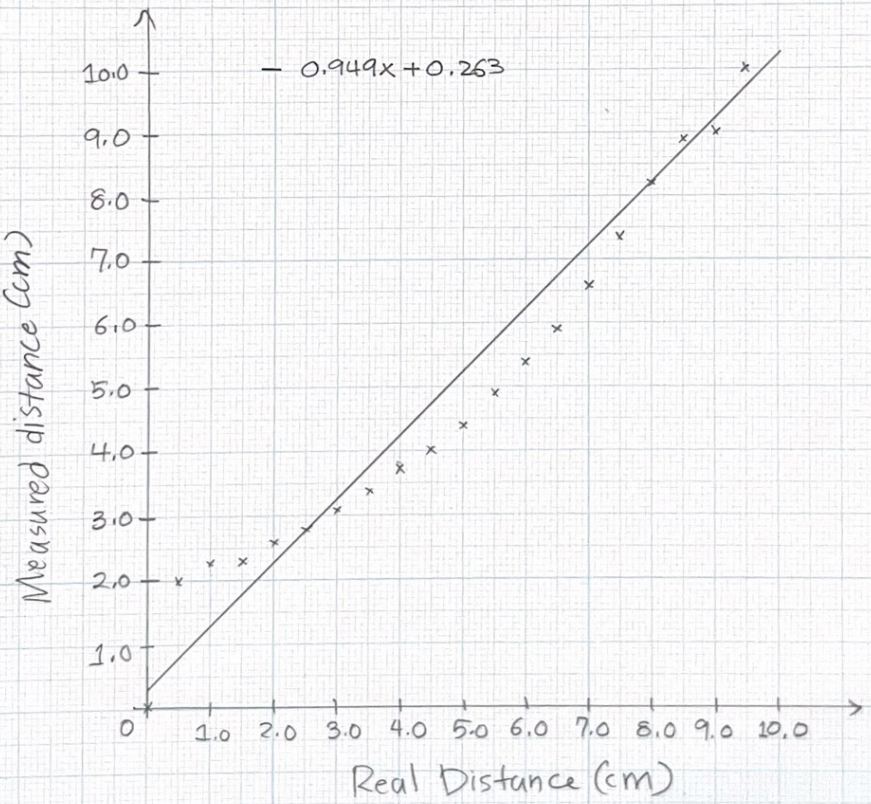
Graph 6: Fig 4-5 Voltage-time data for continuous sin wave input



Graph 7: Fig 4-5 Voltage-time data for triangular wave input.



Graph 8: Distance-Voltage Characteristic Graph



Graph 9: Measured distance vs. Real Distance